DEVELOPMENT OF A WEB-BASED VOTING MANAGEMENT SYSTEM

SUBMITTED BY

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2026

DECLARATION

I, LUBEGA JOEL, confirm that the information contained in this project proposal is primarily based on my own efforts and findings in the field. All the information provided is generally derived from my research investigation.

All sources of information have been duly acknowledged through citations and references according to standard academic guidelines.

Signature: _____

Reg No: 2023-B071-21372

Date:_____

ACKNOWLEDGEMENT

I would like to express my gratitude to the following individuals for their support, guidance, and encouragement throughout my dissertation journey:

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I am thankful to all individuals who participated in the project, sharing their time, knowledge, and experiences. Their contributions were invaluable to this research.

May God bless you all.

APPROVAL

This research project has been submitted to the faculty of Science of Uganda Martyrs University, Department of Computer Science and Information Technology for Examination with the approval of my supervisor.

Signature:…….

Date:……..

LIST OF ABBREVIATIONS

AJAX	Asynchronous JavaScript and XML
API	Application Programming Interface
BVVK	Biometric Voter Verification Kit
CSS	Cascading Style Sheets
DR Form	Declaration of Results Form
EC	Electoral Commission (Uganda)
HTML	HyperText Markup Language
ICT	Information and Communications Technology
MVT	Model-View-Template
NRM	National Resistance Movement
NUP	National Unity Platform
SQL	Structured Query Language
UNESCO	United Nations Educational, Scientific and Cultural Organization

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CHAPTER ONE: INTRODUCTION

1.0 Introduction

In this modern world, the way elections are conducted to exercise our democratic rights has become more important. The integrity of electoral processes fundamentally affects how citizens perceive their government and whether they trust democratic institutions. When elections are fake, faith in democracy erodes; when they are transparent and efficient, democratic participation strengthens.

Uganda's democratic journey has been marked by significant milestones and challenges. The 2026 general elections brought to the forefront critical issues with electoral technology when biometric voter verification kits (BVVK) failed across the country, costing over 268.38 billion Ugandan shillings and leaving many voters deprived. These failures were not just technical glitches they represented a crisis of confidence in the electoral process itself.

While national elections receive the most attention, democracy is first experienced by most Ugandans in a different setting: their primary and secondary schools. The election of prefects, head boys, head girls, and student council representatives is often a young person's first encounter with democratic processes. When school elections are marred by manipulation, violence, or opaque counting, students may become cynical about democracy itself. When they witness transparent, fair elections, they develop trust that lasts a lifetime.

This dissertation addresses these challenges by proposing a Web-Based Online Voting System (VOTLY) built using the Django framework. The system is

designed to serve both national and school-level elections, creating a continuity of democratic participation from the classroom to the national stage.

In this introduction, we will deal with the background of the study, the problem statement, the purpose of the study, the objectives, the scope of the study, and its significance.

1.1 Background of the Study

For centuries, voting has been the primary mechanism through which citizens participate in governance. The word "democracy" itself comes from the Greek words "demos" (people) and "kratos" (rule)—rule by the people. Elections are how that rule is exercised.

Traditional voting methods, while time-tested, face significant challenges in the modern era. Manual vote counting is slow and error-prone. Paper ballots can be lost, stolen, or tampered with. Voter queues can stretch for hours, discouraging participation. These challenges are not unique to Uganda but they affect elections worldwide.

According to the International Institute for Democracy and Electoral Assistance (International IDEA), the use of technology in elections has grown significantly over the past two decades. From electronic voter registration to biometric verification and online result transmission, governments and electoral bodies have embraced technology as a means to enhance efficiency, reduce fraud, and increase transparency. However, the successful implementation of electoral technology requires careful consideration of local contexts, infrastructure limitations, and human capacity.

In Uganda, the Electoral Commission has introduced technology into the electoral process. The 2016 elections introduced biometric voter registration. The 2021 elections deployed biometric verification kits for the first time. The 2026 elections represented the most ambitious technological deployment yet, with over 109,000 biometric kits procured at Ugandan shillings over 268.38 billion.

The main objective of this research is to develop a web-based online voting system with a friendly user interface that addresses the limitations of current systems while building on the lessons learned from recent electoral experiences.

1.2 Problem Statement

The current electoral systems in Uganda both national and school-level face critical challenges that undermine democratic participation and trust:

At National Level:

1. Technical Failures: The catastrophic failure of biometric verification kits during the 2026 elections demonstrated that technology alone is insufficient without robust design, thorough testing, and contingency planning.

2. Inefficient Processes: Manual vote counting requires thousands of officials to count ballots by hand across 34,000+ polling stations, delaying results for days.

3. High Operational Costs: The 2026 election cost over UGX 1.3 trillion, a burden on Uganda's economy.

4. Limited Voter Access: Many Ugandans face barriers to physical voting, including those in remote areas, persons with disabilities, and urban workers who cannot take a full day off.

5. Security Vulnerabilities: Manual systems are susceptible to ballot stuffing, multiple voting, and result manipulation.

6. Lack of Real-Time Transparency: Citizens cannot observe the voting process in real-time, reducing trust in declared results.

At School Level:

1. Educational Disruption: School elections consume valuable instructional time, disrupting learning for all students.

2. Ballot Security: Improvised ballot boxes are easily tampered with.

3. Multiple Voting: Without reliable voter rolls, students vote multiple times.

4. Counting Errors and Delays: Manual counting leads to errors and delays.

5. Negative Civic Lessons: When students witness rigged elections, they learn that democracy is unreliable.

1.3 Purpose of the Study

The purpose of this platform is to provide a secure, efficient, and user-friendly web-based online voting system that addresses the electoral challenges facing Uganda at both national and school levels. The system will:

1.4 Objectives

1.4.1 Main Objective

To design and implement a secure, efficient, and user-friendly Web-Based Online Voting System using Django that serves both national and school-level elections in Uganda.

1.4.2 Specific Objectives

i. To analyze the current electoral systems in Uganda at both national and school levels, identifying their strengths, weaknesses, and specific requirements through fact-finding methods including questionnaires, interviews, and observation.

ii. To design a Django-based online voting system architecture incorporating the Model-View-Template pattern, with multi-tenancy support for schools, three main modules (Account, Voting, Administrator), and scalability for elections of various sizes.

iii. To develop the system using Python, Django, and Bootstrap, implementing core features including user authentication, ballot generation, vote submission, and automated result tabulation.

iv. To test and validate the system's functionality, security, and usability through unit testing, integration testing, and user acceptance testing with representatives.

1.5 Research Scope

1.5.1 Geographic Scope

This research will take place in Uganda, focusing on electoral processes at both national level and in selected secondary schools across various districts including Kampala, Wakiso. The platform will be designed for use by Ugandan voters and school communities.

1.5.2 Time Scope

The research will take place over a period of 4 months.

1.5.3 Functional Scope

The platform will provide the following services:

For National Elections:

- Voter registration and authentication
- Dynamic ballot display based on positions
- Secure vote casting
- Real-time result tabulation
- Administrator dashboard for election management

For School Elections:

- All national features, plus:
- School-specific administration
- Student voter management
- Minimal disruption to school schedule

1.6 Significance of the Study

This study is important because it demonstrates how technology can improve the voting process by making it more efficient, transparent, and reliable. An online voting system reduces the need for manual ballot counting, minimizes human errors, and speeds up the announcement of election results. For national democracy, such a system can lower the high costs associated with traditional elections and improve public confidence in the electoral process through faster and more transparent result management.

The system is also beneficial in educational institutions, particularly in schools where elections are often conducted manually. By introducing an online voting platform, schools can conduct elections quickly without interrupting academic activities. It also helps students understand modern democratic processes and encourages participation in leadership activities, thereby strengthening democratic values from an early stage.

Furthermore, the system benefits voters, administrators, and researchers. Voters can conveniently cast their votes from any location and view results instantly. Electoral bodies and school administrators can monitor the election process in real time and reduce the workload involved in organizing elections. In addition, the system can serve as a foundation for future researchers who may wish to improve or expand online voting technologies

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

In an age characterized by rapid technological advancements and an increasing reliance on the internet for information and services, online voting systems have emerged as pivotal tools in the modernization of democratic processes worldwide. This literature review aims to provide an overview of key themes and developments in the field of online voting systems, with particular attention to the Ugandan context.

In this section, first we will analyze the background of online voting systems and their features, then we review the evolution of electronic voting platforms, Uganda's electoral technology journey, democracy in schools, and other existing platforms on the web that are related to our proposed Online Voting System.

2.1 Background of Online Voting Systems

Online voting systems are platforms that enable voters to cast their ballots via the internet from any location with an internet connection. These systems save voters time, increase convenience, and can potentially increase voter participation (Alvarez, R.M. and Hall, T.E., 2018).

One of the primary benefits of online voting systems is their potential to increase voter turnout. Research indicates that convenience plays a significant role in whether citizens vote. According to studies by the International Institute for Democracy and Electoral Assistance(IIDEA), countries that have implemented online voting have seen increased participation, particularly among younger voters and those living abroad.

However, the security and trustworthiness of online voting systems remain primary concerns. Research by Jones et al. (2020) highlights the vulnerabilities inherent in

internet-based systems, including the risks of hacking, denial-of-service attacks, and vote manipulation. This underscores the need for robust security mechanisms, including encryption, multi-factor authentication, and comprehensive audit trails.

The integration of multi-factor authentication, particularly through mobile phone verification, has emerged as a significant trend in online voting systems. OTP (One-Time Password) verification via SMS provides an additional layer of security beyond simple username and password combinations.

2.2 The Evolution and Development of Electronic Voting Platforms

The evolution of voting technology has transformed the way elections are conducted, moving from traditional paper based elections to advanced electronic voting solutions. Over time, election technology has incorporated innovations such as mechanical and computerized voting machines, ultimately leading to the development of modern electronic voting systems.

These advancements have not only increased the efficiency and accessibility of elections but also raised important concerns about electronic voting security. Today, as we continue to embrace digital tools for elections, ensuring the integrity and trustworthiness of these systems remains a critical focus.

The Era of Paper-Based Voting

Paper-based voting has been the foundation of electoral processes for centuries. This system relied on physical ballots, manual counting, and human oversight. While paper-based voting ensured a tangible and auditable record of votes, it was

often time-consuming, prone to errors, and limited in terms of accessibility and efficiency.

The Advent of Mechanical Voting Machines

The introduction of mechanical voting machines in the 19th and 20th centuries brought automation and improved accuracy to the voting process.

These machines facilitated the recording and tabulation of votes, reducing counting errors and expediting the electoral process. However, mechanical voting machines still relied on physical mechanisms and lacked the flexibility and advanced features of modern electronic voting systems.

The Rise of Computerized Voting Systems

Computerized voting systems marked a significant leap forward in voting technology. These systems utilized digital technology, allowing for faster tabulation, increased efficiency, and improved accuracy.

Optical scan systems and direct-recording electronic (DRE) voting machines became common, offering features such as vote verification, automated counting, and accessibility enhancements. However, concerns regarding security and auditability arose with the adoption of these systems.

Modern Electronic Voting Solutions

Modern electronic voting solutions represent the latest stage in the evolution of voting technology. These systems leverage advanced features such as touch-screen interfaces, biometrics, encryption, and secure transmission protocols.

Electronic voting ensures faster results, accessibility for diverse populations, and enhanced accuracy. While security and privacy concerns are critical considerations, robust measures are in place to address these challenges and uphold the integrity of the electoral process.

2.3 Uganda's Electoral Technology Journey

2.3.1 Pre-Independence to 2001: Manual Voting

From the 1962 independence elections through 2001, Uganda relied entirely on manual voting systems. Voters registered on paper, were verified visually, and cast paper ballots counted by hand. While simple and understandable, this system was susceptible to multiple voting, ballot stuffing, and slow counting.

2.3.2 2006-2011: Introduction of Voter Registration Technology

The 2006 elections introduced computerized voter registers, though verification remained largely manual. The 2011 elections saw improved voter lists but continued manual verification and voting.

2.3.3 2016-2021: Biometric Registration and Verification

The 2016 elections introduced biometric voter registration, capturing fingerprints and photographs. The 2021 elections deployed biometric verification kits at polling stations for the first time, though numerous technical glitches were reported.

2.3.4 2026 Elections: Biometric Verification Kits and Challenges

The 2026 elections represented Uganda's most ambitious technological deployment. The Electoral Commission procured 109,142 biometric verification kits at a cost of UGX 268.38 billion. Each kit featured fingerprint scanners, facial recognition cameras, solar charging capability, and battery backup.

However, widespread failures occurred on election day:

- Machines failed to activate at numerous polling stations
- Some polling stations received no machines at all
- Even President Museveni experienced fingerprint verification failure
- Machines froze, rejected legitimate voters, and displayed errors
- Election officials reverted to manual registers after public pressure

Analysts identified multiple contributing factors:

- Inadequate Testing; Systems were not thoroughly tested in real-world conditions
- Poor Training; Operators received training only little period before the election
- Environmental Factors; poor Internet, Dust, heat, and humidity affected performance

2.4 Democracy and Elections in Schools

Research consistently demonstrates that school elections play a crucial role in democratic formation. According to the International Association for the Evaluation of Educational Achievement, "schools are the primary institutions for

civic socialization outside the family." Students who participate in school elections are more likely to vote as adults.

The Center for Information and Research on Civic Learning and Engagement finds that "experiences with democratic processes in school, including student government elections, increase young people's sense that they can influence political processes."

2.4.2 Current Practices in School Elections

School elections in Uganda typically follow patterns established during colonial times. According to a 2023 study by the Uganda National Examinations Board (UNEB), approximately 65% of secondary schools hold elections for student leadership, though only 40% have formal electoral guidelines.

2.5 Existing Related Platforms

2.5.1 The Kunga App (Uganda)

Launched by National Unity Platform (NUP) leader Robert Kyagulanyi (Bobi Wine) ahead of the 2026 elections, the Kunga App is a decentralized monitoring tool for volunteers to document and report real-time events at polling stations. Key features include evidence collection through smartphone cameras, DR form verification through photo uploads, offline functionality, and anonymity protections.

Relevance: The Kunga App demonstrates the power of decentralized, citizen-driven election monitoring and the importance of designing for Uganda's connectivity realities.

2.5.2 ONC Vote Protection and Monitoring System

The Office of the National Chairman unveiled a digital Vote Protection and Monitoring System in November 2025, featuring real-time voter turnout monitoring, geolocation tracking of vote protectors, suspicious movement flagging, and field operative identity verification.

Relevance: This system demonstrates the potential of integrated data platforms for election monitoring and coordination.

2.5.3 Kenya's Electronic Voting Experience

Kenya's electoral technology journey offers valuable lessons. The 2013 elections introduced biometric voter registration and electronic result transmission. The 2017 elections saw the Supreme Court nullify presidential results partly due to "irregularities and illegalities" in the transmission system. The 2022 elections improved systems with better transparency and parallel counting.

Lessons: Technology must be transparent and auditable; stakeholder consensus is essential; redundancy and fallback mechanisms are critical.

2.5.4 Estonia's Internet Voting System

Estonia has conducted internet voting since 2005, with over 50% of votes cast online in recent elections. Key features include national ID cards with digital signatures, multiple layers of encryption, "one vote, one voter" enforcement, ability to change votes during the voting period, and paper audit trail options.

Relevance: Estonia demonstrates that secure internet voting is possible with proper design and infrastructure.

Conclusion

Online voting platforms help modernize democratic processes, increase participation, and improve efficiency. Digital platforms are helping to expand democratic participation into areas where voters may have difficulty accessing physical polling stations, providing added convenience for those living busy lives and in need of accessible voting solutions.

However, the Ugandan experience with biometric technology in 2026 demonstrates that technology must be carefully designed, thoroughly tested, and implemented with adequate training and contingency planning. This project builds on these lessons to create a robust online voting system suitable for Uganda's context.

CHAPTER THREE: RESEARCH METHODOLOGY

3.0 Introduction

The idea behind developing the Online Voting System is to provide users with a platform that enables secure, efficient, and transparent voting for both national and school-level elections. The system leverages Django's MVT architecture to create a robust, scalable solution that addresses the challenges identified in Uganda's current electoral processes.

This chapter explains the methods and processes that will be used to achieve the stated objectives of the study. It involves all the step tasks that will be carried out while collecting information with the aim of designing an online voting system. The methodology describes the research design, the case study, the requirement collection tools to be used, and phases involved in development of the platform.

3.1 Research Methodology

According to Sujata Mehta (2023), research methodology refers to the systematic and scientific approach employed to collect, analyze, and interpret data for the purpose of answering research questions or testing hypotheses.

For this research, a mixed-methods approach will be used, combining quantitative data from questionnaires with qualitative insights from interviews and observations. This approach ensures a comprehensive understanding of the requirements and challenges.

3.2 Research Design

Bhat, Adi (2023) argues that research design is a framework of research methods and techniques chosen by a researcher to conduct a study. The design allows researchers to sharpen the research methods suitable for the subject matter and set up their studies for success.

This research will use a case study design, focusing on Uganda's electoral processes at both national and school levels. The design focuses on the tools of data collection and data analysis.

3.3 Data Collection Methods

The primary data collection method was a questionnaire-based interview. The secondary data collection was collected from books and online documents.

3.4 Creation of Platform

For this research, I have used the agile methodology based on iterative development. The steps are as follows:

1. Requirements Gathering: Collect and document requirements from stakeholders

2. Design: Create system architecture, database design, and user interface mockups

3. Development: Build the platform using Django framework

4. Testing: Conduct unit, integration, and user acceptance testing

5. Deployment: Deploy the platform for pilot testing

6. Evaluation: Gather feedback and make improvements

3.6 Technology and Tools Used

For the development of the platform, the technology and tools used are:

i. Python Programming Language: Backend development and business logic

ii. Django Web Framework: MVT architecture, built-in authentication, ORM, and admin interface

iii. SQLite Database: Development database (PostgreSQL for production)

iv. HTML, CSS, and Bootstrap: Frontend structure, design, and responsive interface

v. JavaScript and AJAX: Dynamic content loading and interactive features

vi. Visual Studio Code: Code editor for development

vii. Git: Version control

viii. Web Browsers: Chrome, Firefox for testing and implementation

3.7 Platform Implementation

The platform will be implemented using the following:

Django

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. It will be used because it is secure, scalable, and includes built-in features for authentication, admin interfaces, and database management.

SQLite/PostgreSQL

SQLite will be used for development, with PostgreSQL for production deployment. These databases provide reliable, ACID-compliant storage for voting data.

HTML, CSS, Bootstrap

These will be used as the web template to design interfaces because:

- They are easy to use and learn

- They provide responsive design for mobile and desktop

- Bootstrap ensures consistent, professional appearance

CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

4.0 Introduction

The Online Voting System is a web-based platform that has the objectives of enabling secure, efficient, and transparent voting for both national and school-level elections in Uganda. This documentation provides a description of the features that have been embedded in the platform in order to achieve these objectives.

4.1 Platform Requirements

4.1.1 Voter-Facing Functional Requirements

i. The platform shall allow new voters to register by providing necessary personal and contact information.

ii. The platform shall validate and verify the registration details to ensure accuracy and prevent duplicate entries.

iii. Registered voters shall be able to log in to the portal using their credentials.

iv. The platform shall display all positions for the election with corresponding candidates.

v. The platform shall display all positions for the election with corresponding candidates.

vi. .Before final submission, the platform shall display a preview of the voter's selections for confirmation.

vii. Upon confirmation, the platform shall record the votes and mark the voter as having voted

4.1.2 Administrator-Facing Functional Requirements

i. Administrators shall be able to log in securely to the admin panel.
ii. The platform shall allow administrators to manage voters (add, edit, delete, bulk upload).
iii. Administrators shall be able to manage candidates (add, edit, delete, assign to positions).
iv. The platform shall allow creation and management of positions (title, description, max votes).
v. The platform shall display real-time results and turnout statistics.
vi. Administrators shall be able to generate and export result reports

4.1.3 Usability Requirements

i. The user interface shall be intuitive and clean.
ii. The platform shall be mobile-responsive (works on smartphones.
iii. Navigation shall be consistent across all pages.
iv. Error messages shall be clear and helpful.

4.1.4 Accessibility Requirements

i. The platform shall be compatible with screen readers.
ii. Text shall be resizable.
iii. Sufficient color contrast shall be maintained.
iv. The platform shall be keyboard navigable.

4.1.5 Software Requirements

The platform requires compatibility such that it is able to run and perform its required tasks properly. The following are the software requirements:

Item	Minimum Requirement
Operating System	Windows 10/11, Linux, macOS
Web Server	Django development server / Gunicorn / Apache
Database	SQLite (dev), PostgreSQL (production)
Python	Python 3.9 or higher
Django	Django 4.2 or higher
Browser	Google Chrome, Mozilla Firefox, Safari, Microsoft Edge

4.1.6 Hardware Requirements

The platform requirements for running and performing its required tasks properly should be on hardware facilities. The following are the hardware requirements:

Item	Minimum Requirement
Processor	Intel Core i3 / AMD equivalent (2.0 GHz)
Memory (RAM)	4GB (development), 8GB+ (production server)
Storage Space	20GB for application and database
Network	Internet connection (for users), Broadband for server
Client Device	Any device with web browser and internet connection

4.2 Platform Design

4.2.1 System Architecture: Django MVT

The proposed online voting platform is built using the Model–View–Template (MVT) architecture by Django. This structure separates the application into different components responsible for managing data, processing logic, and displaying information to users. Dividing these components enhances system maintainability, scalability, and security.

User Browser (Client)

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▼

URL Configuration (urls.py)

|

▼

Views (Application Logic)

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▼

Models (Data Handling)

|

▼

Database (SQLite/PostgreSQL)

▲

|

Templates (User Interface)

|

▼

User Response

4.2.2 Database Design (ERD)

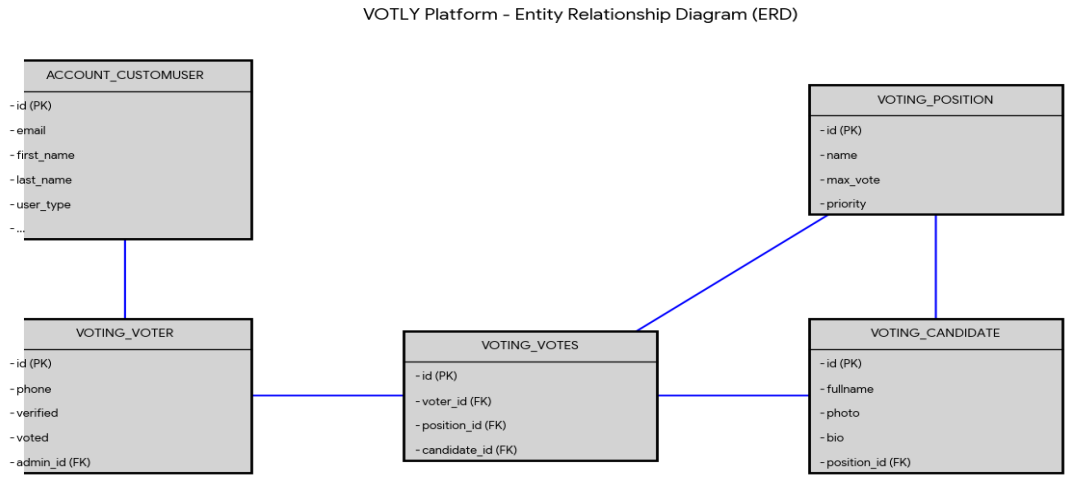
CustomUser (account_customuser): The base identity for all users in the system, storing credentials and user types.

Voter (voting_voter): Extends the user functionality, linked to CustomUser via admin_id. It tracks whether a person has voted and their verified status.

Position (voting_position): Represents the roles available for election (e.g., President, Speaker). It defines the max_vote allowed for that category.

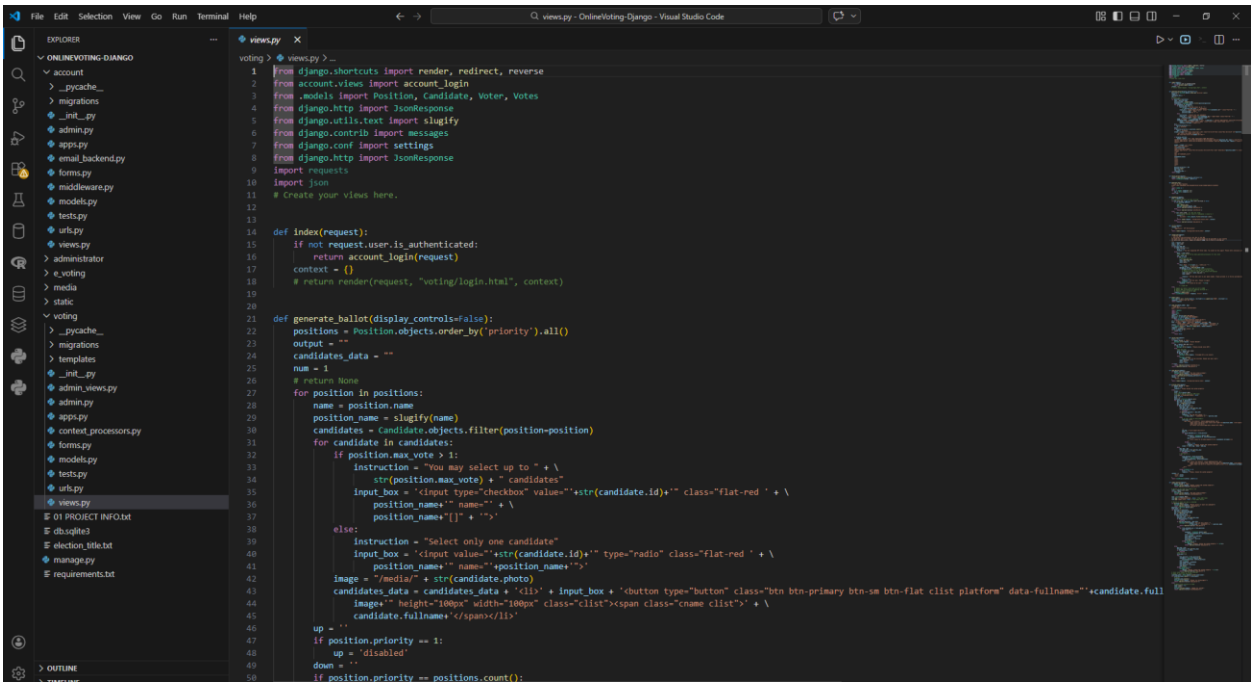
Candidate (voting_candidate): Individuals running for a specific position_id. It stores their fullname, bio, and photo.

Votes (voting_votes): The central transaction table that connects **a voter_id to a candidate_id and the specific position_id they are voting for.**

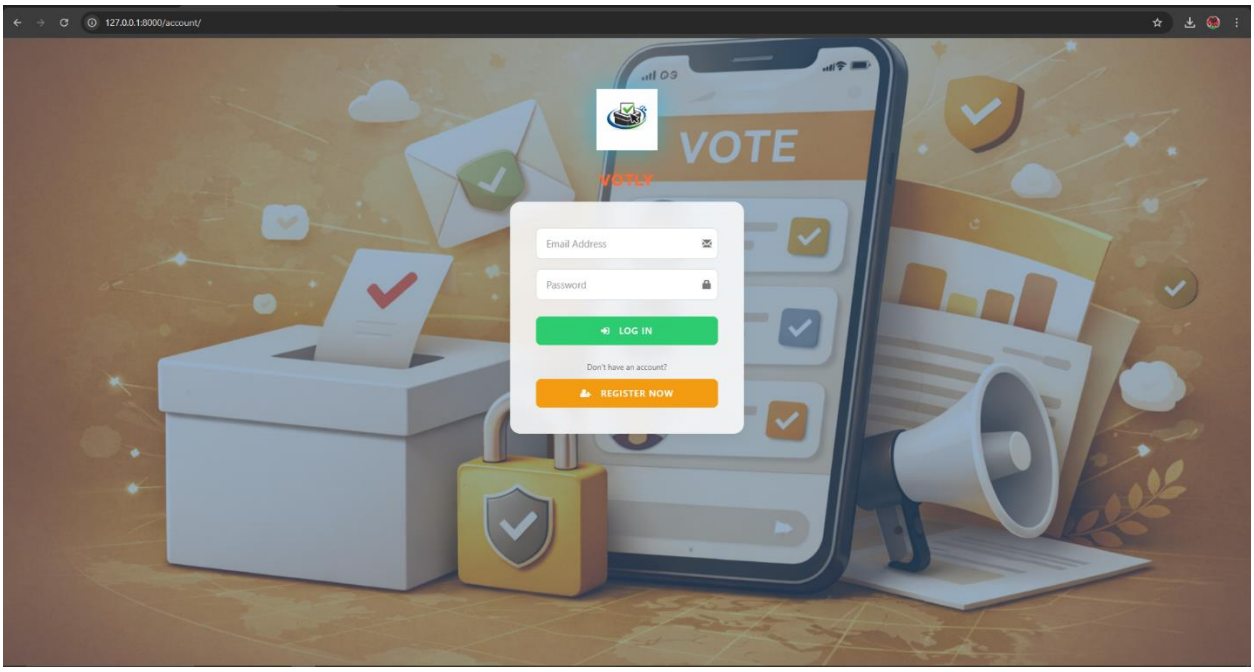


4.3 Platform Implementation

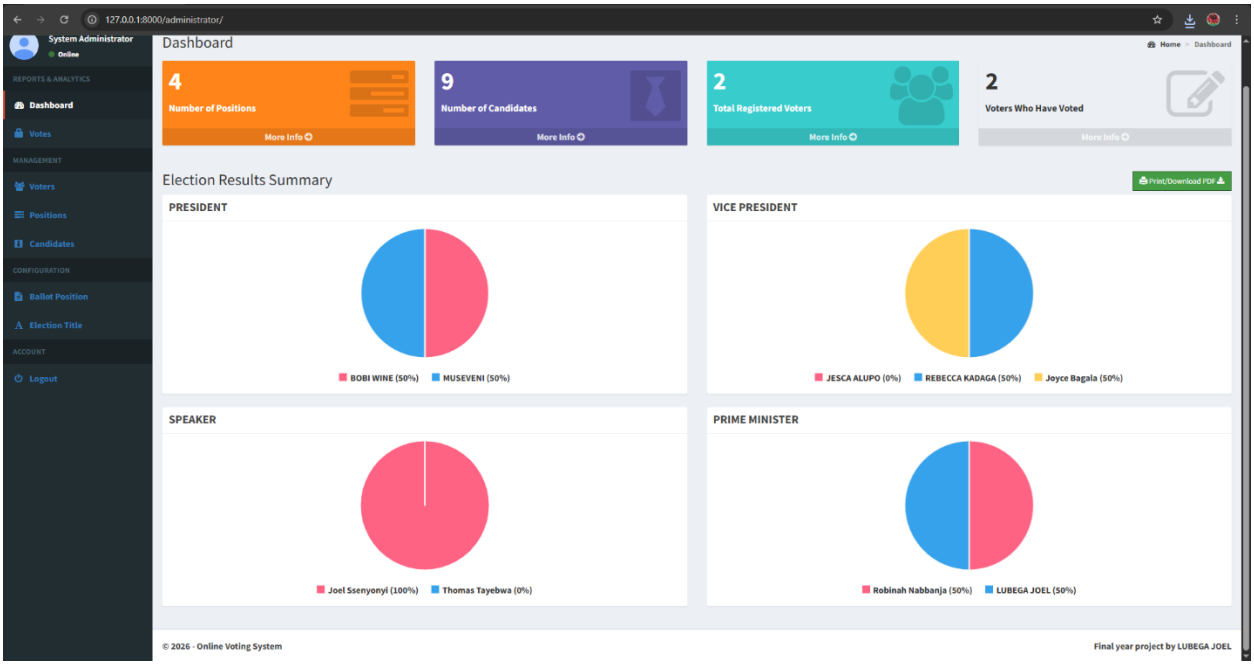
This section looks at the procedure used to implement the user interface. Data entry interfaces were designed to be as simple and functional as possible to make the system easier to use. The proposed system was developed using Django framework with Python, HTML, CSS, and Bootstrap.



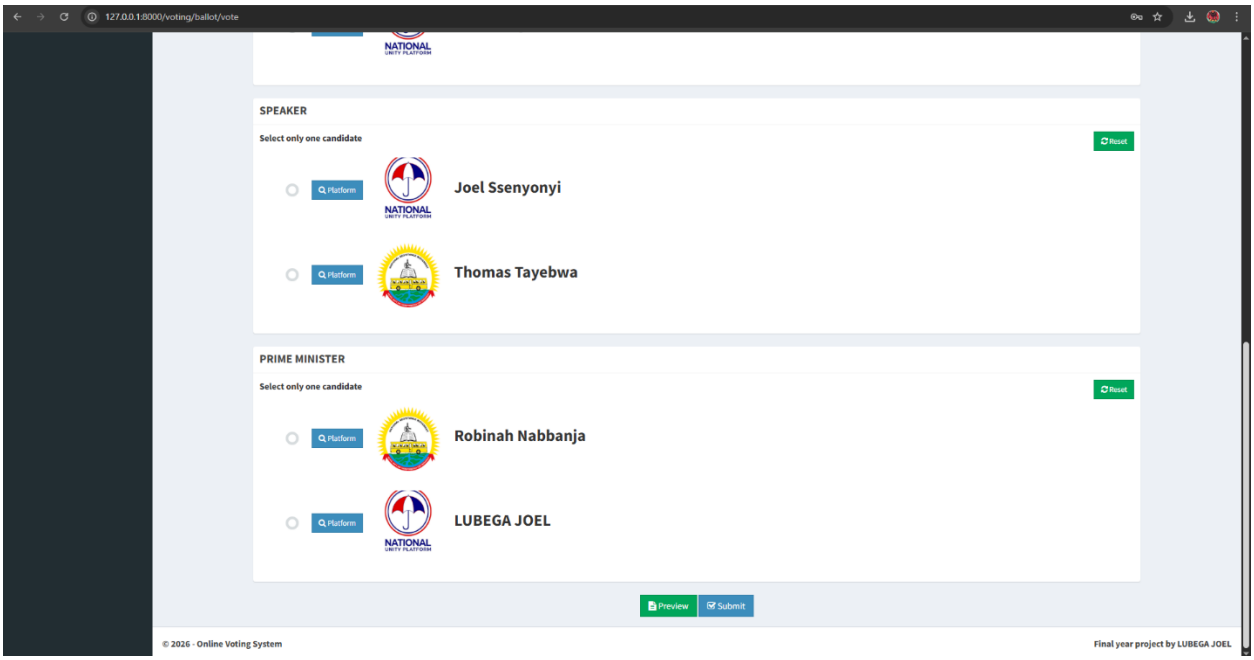
Homepage and User Authentication



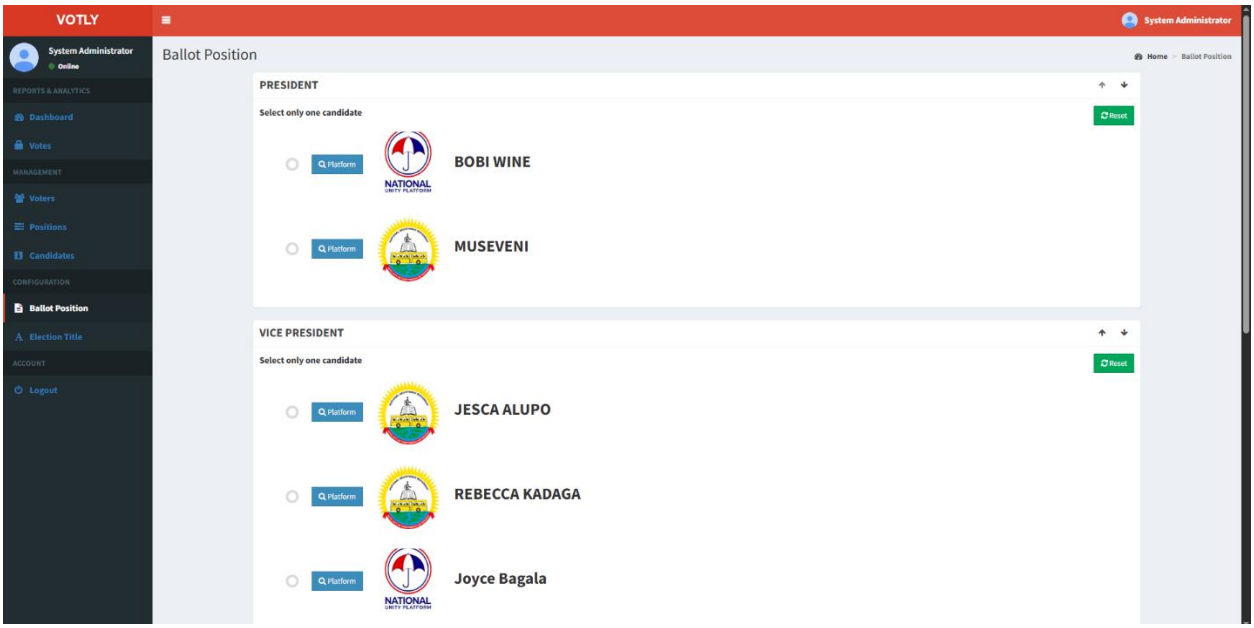
User Interface of the admin



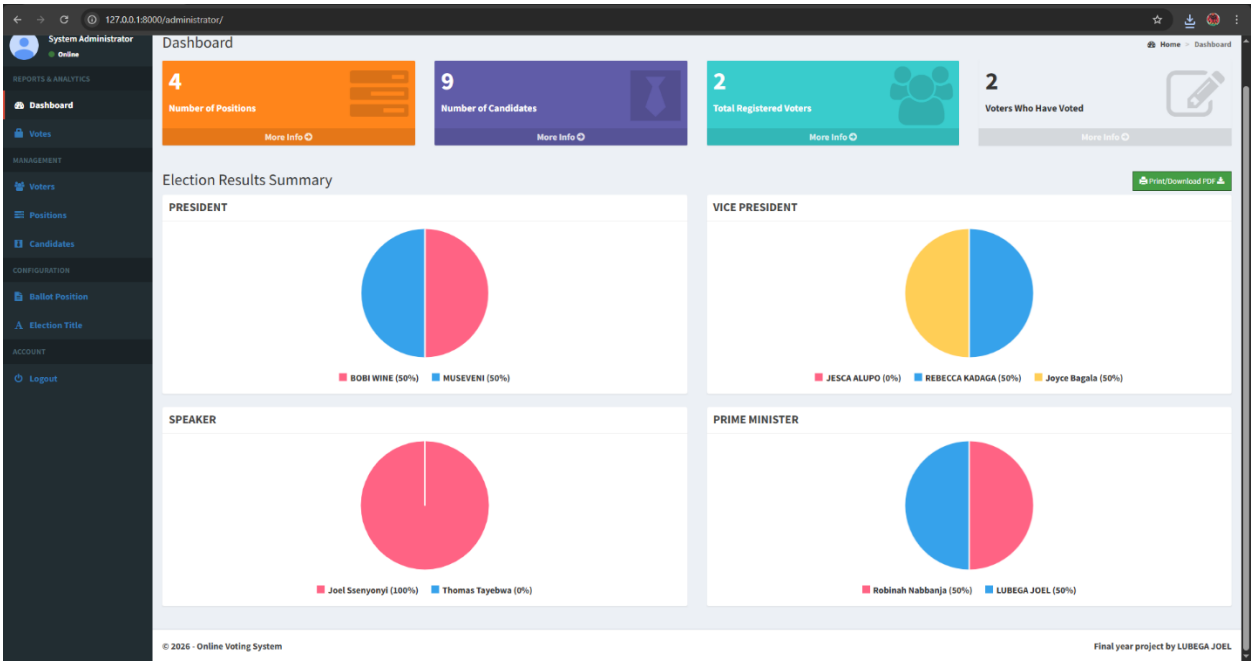
User Interface of the voter



Position Management Page



Results summary Display



4.4 Platform Testing

The testing was done through exposing the platform to users in an operational environment and checking the results. Application programs were executed with the intent of finding errors and confirming that functional and user most requirements have been met.

Unit Testing

Each module was tested separately to ensure individual components function correctly. Tests included:

- Authentication module testing

- Ballot generation testing
- Vote submission testing
- Result output testing

Integration Testing

After unit testing, an overall test was conducted with all modules integrated into the platform. This tested the interactions between components and the overall system workflow.

Acceptance Testing

End users of the platform, including fellow students, tested the applications to confirm that their requirements were met. Feedback was collected and used for refinements.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.0 Introduction

This final chapter notes the conclusion of the study and recommendations from the researcher to anyone reading the research and to those who may do research on a similar topic.

5.1 Conclusion

The Online Voting System is an example of a digital innovation that can have a positive impact on democratic participation and electoral efficiency in Uganda. In this study, we have proposed a web-based online voting system built with Django that serves both national and school-level elections.

The uniqueness of this system lies in its features:

- Django MVT Architecture: Providing a secure, scalable foundation

- Multi-Tenancy Support: Serving both national and school elections from a single codebase

- Real-Time Results: Eliminating delays in result declaration

- User-Friendly Interfaces: Ensuring accessibility for all users

- Comprehensive Audit Trails: Enabling post-election verification

The system addresses the critical weaknesses exposed during the 2026 Uganda elections, where biometric verification kits failed catastrophically. By learning from those failures and incorporating robust testing, user-centered design, and clear contingency planning, this system offers a pathway to more reliable electoral technology.

For schools, the system transforms elections from disruptive, error-prone events into efficient, positive learning experiences that build democratic values rather than undermining them.

5.2 Recommendations

5.2.1 Recommendations for the Electoral Commission

i. The Electoral Commission should consider piloting this online voting system in selected constituencies or for specific elections (such as diaspora voting) to evaluate its effectiveness in the Ugandan context.

ii. The Commission should invest in comprehensive voter education to build trust in online voting and ensure all citizens understand how to use the system.

iii. Any deployment of online voting should include robust contingency plans, including manual backup procedures, as the 2026 experience demonstrated that technology can fail.

iv. The Commission should work with Parliament to update electoral laws to explicitly provide for online voting, ensuring a clear legal framework.

v. The Commission should prioritize transparency in system design, allowing independent security audits and stakeholder review.

5.2.2 Recommendations for the Ministry of Education and Sports

i. The Ministry should develop national guidelines for electronic student elections, providing schools with a clear framework for adopting online voting.

ii. The Ministry should consider including online voting in the civic education curriculum, teaching students about both democratic processes and the technology that enables them.

iii. Schools should be encouraged to adopt online voting to reduce class disruption and provide positive civic education experiences.

iv. The Ministry should work with telecommunications companies to ensure affordable internet access for schools, particularly in rural areas.

5.2.3 Recommendations for School Administrators

i. Schools should consider adopting online voting for student elections to save time, reduce costs, and provide students with positive democratic experiences.

ii. Administrators should ensure all students have access to voting devices, whether through personal smartphones or school computer labs.

iii. Schools should use online voting as a teaching opportunity, discussing the technology and its role in modern democracy.

iv. Parent-teacher associations should be informed about the system to build trust and support.

5.2.4 Recommendations for Future Researchers

i. Future researchers should consider integrating biometric authentication with the existing OTP system for enhanced security.

ii. Research on blockchain-based vote validation could provide additional transparency and auditability.

iii. Studies on the long-term impact of online voting on democratic participation in Uganda would be valuable.

iv. Comparative studies between different online voting platforms and their effectiveness in African contexts would contribute to the body of knowledge.

5.3 Future Enhancements

There are areas which can be enhanced and improved to make the platform better:

5.3.1 Biometric Authentication Integration

Integrating fingerprint or facial recognition with the existing system would provide multi-factor authentication that combines something the voter knows (password), something the voter has (phone), and something the voter is (biometric).

5.3.2 Blockchain-Based Vote Validation

Blockchain technology could provide an immutable, transparent ledger of votes that anyone can verify while maintaining voter anonymity. This would address concerns about auditability and trust.

5.3.3 Multi-Language Support

Adding support for major Ugandan languages (Luganda, Acholi, Runyankole, Lusoga, etc.) would make the platform accessible to more citizens, particularly in rural areas.

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Appendix

Questionnaire for Students (School Elections)

ONLINE VOTING SYSTEM - STUDENT QUESTIONNAIRE

Dear Student,

I am conducting research on the development of an Online Voting System for school elections in Uganda. Your participation in this survey is greatly appreciated. All information provided will be used for academic purposes only.

Section A: Demographic Information

1. Age: _____

2. Gender:

- [ ] Male
- [ ] Female

3. School Name: _____

4. Class/Form: _____

Section B: School Election Experience

5. Have you ever voted in school elections (prefect, head boy/girl, etc.)?

- [ ] Yes
- [ ] No

6. How would you rate your school election experience?

- [ ] Excellent
- [ ] Good
- [ ] Fair
- [ ] Poor

7. Did voting disrupt your classes?

- [ ] Yes, significantly
- [ ] Yes, somewhat
- [ ] No, not really

8. How long did it take to get election results?

- [ ] Same day
- [ ] Next day

- [ ] Two or more days later

9. Have you ever witnessed or suspected any problems in school elections?

- [ ] Yes (please specify: _____)
- [ ] No

Section C: Technology Access

10. Do you own a mobile phone?

- [ ] Yes
- [ ] No

11. If yes, is it a smartphone?

- [ ] Yes
- [ ] No

12. Do you have access to the internet at school?

- [ ] Yes
- [ ] No
- [ ] Sometimes

Section D: Online Voting Attitudes

13. Would you trust an online voting system for school elections?

- [ ] Yes, definitely
- [ ] Yes, probably
- [ ] Not sure
- [ ] No

14. Would you prefer to get election results immediately after voting?

- [ ] Yes
- [ ] No
- [ ] Doesn't matter

15. Any comments about how school elections could be improved?

_____

_____

Thank you for your participation!

APPENDIX B:

User Manual

VOTLY VOTING SYSTEM - USER MANUAL

1. Introduction

Welcome to the Online Voting System user manual. This guide will help you understand how to use the platform for voting in elections.

2. System Requirements

- A device with internet connection (computer, smartphone, tablet)

- Modern web browser (Chrome, Firefox, Safari, Edge)

3. Getting Started

3.1 Registration

1. Navigate to the Online Voting System homepage

2. Click on "Register" button

3. Fill in the required information:

   - Full name

   - Phone number

   - Email address

   - Password

4. Click "Register"

3.2 Login

1. Click on "Login" button

2. Enter your email and password

3. Click "Login"

4. Voting Process

4.1 Accessing the Ballot

1. Click on "Cast Vote" or "View Ballot"

2. The ballot will display all positions for the current election

4.2 Making Your Selections

1. For each position, you will see a list of candidates

2. Select your preferred candidate(s) by clicking on the checkbox or radio button

3. Note the maximum number of votes allowed per position

4. You can change your selections at any time before submission

4.3 Previewing Your Vote

1. After making all selections, click "Preview Vote"

2. Review your selections carefully

3. If corrections are needed, click "Back to Ballot"

4. If satisfied, click "Submit Vote"

4.4 Login Issues

- Verify email and password

- Contact administrator for assistance

4.5 Voting Issues

- Ensure you have not already voted (one vote per voter)

- Check that election is active

- Clear browser cache and try again

5. Security Tips

- Never share your password

- Log out after completing your vote

- Use a secure internet connection

- Report any suspicious activity to administrators